

H2 Standard integrals

Name: _____

Class: _____

Date: _____

Time: **375 minutes**

Marks: **322 marks**

Comments:

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Q1.

Given that $\frac{dy}{dx} = \frac{1}{6x^2}$ find y

Circle your answer.

$$\frac{-1}{3x^3} + c$$

$$\frac{1}{2x^3} + c$$

$$\frac{-1}{6x} + c$$

$$\frac{-1}{3x} + c$$

(Total 1 mark)

Q2.

$$f'(x) = \left(2x - \frac{3}{x}\right)^2 \text{ and } f(3) = 2$$

Find $f(x)$.

(Total 4 marks)

Q3.

- (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\frac{1}{\sqrt{4+x}}$$

(3)

- (b) Hence, find the first three terms of the binomial expansion of $\frac{1}{\sqrt{4-x^3}}$

(2)

- (c) Using your answer to part (b), find an approximation for $\int_0^1 \frac{1}{\sqrt{4-x^3}} dx$, giving your answer to seven decimal places.

(3)

- (d) (i) Edward, a student, decides to use this method to find a more accurate value for the integral by increasing the number of terms of the binomial expansion used.

Explain clearly whether Edward's approximation will be an overestimate, an underestimate, or if it is impossible to tell.

(2)

- (ii) Edward goes on to use the expansion from part (b) to find an approximation

$$\text{for } \int_{-2}^0 \frac{1}{\sqrt{4-x^3}} dx$$

Explain why Edward's approximation is invalid.

(2)

(Total 12 marks)

Q4.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

(a) Find an expression, in terms of t , for the force acting on the particle.

(3)

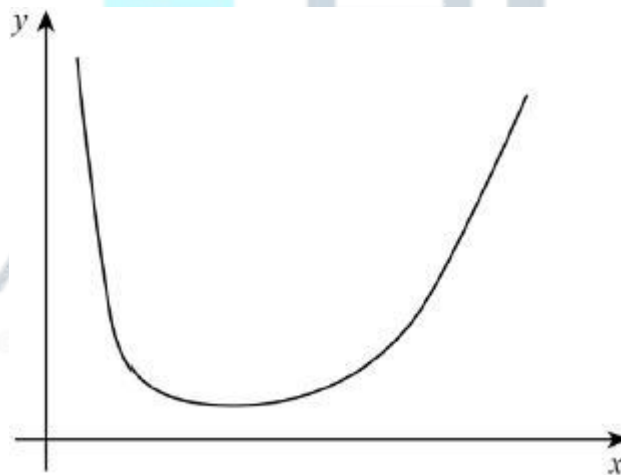
(b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)

Q5.

A curve has equation $y = 6x^2 + \frac{8}{x^2}$ and is sketched below for $x > 0$



Find the area of the region bounded by the curve, the x -axis and the lines $x = a$ and $x = 2a$, where $a > 0$, giving your answer in terms of a

(Total 4 marks)

Q6.

The height, x metres, of a column of water in a fountain display satisfies the differential

equation $\frac{dx}{dt} = \frac{8 \sin 2t}{3\sqrt{x}}$, where t is the time in seconds after the display begins.

(a) Solve the differential equation, given that initially the column of water has zero height.

Express your answer in the form $x = f(t)$

(7)

- (b) Find the maximum height of the column of water, giving your answer to the nearest cm.

(1)

(Total 8 marks)

Q7.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally. The velocity, $\mathbf{v} \text{ m s}^{-1}$, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that the parachutist is at the origin when $t = 0$

Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t .

(4)

- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)

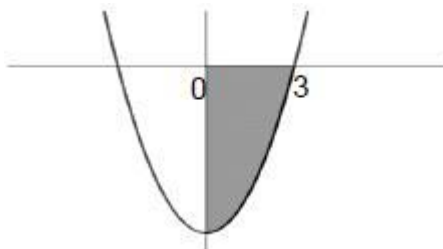
(Total 11 marks)

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Q8.

The graph of $y = x^2 - 9$ is shown below.



Find the area of the shaded region.

Circle your answer.

-18

-6

6

18

(Total 1 mark)

Q9.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A .

- (a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

- (b) When it was first noticed, the weed covered an area of 0.25 m^2 . Twenty days later the weed covered an area of 0.5 m^2

- (i) State the value of B .

(1)

- (ii) Show that the the model for the area covered by the weed can be written as

$$A = 2^{20^{-2}}$$

(4)

- (iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

- (c) State one limitation of the model.

(1)

- (d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

Q10.

$\int_1^2 x^3 \ln(2x) dx$ can be written in the form $p \ln 2 + q$, where p and q are rational numbers.

Find p and q .

(Total 5 marks)

Q11.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_1^5 \frac{1}{x^2 + 1} dx$$

giving your answer to three significant figures.

(4)

- (b) (i) Find $\int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$, giving the coefficient of each term in its simplest form.

(3)

- (ii) Hence find the value of $\int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$.

(2)

(Total 9 marks)

Q12.

- (a) Show that

$$\int_0^{\ln 2} e^{1-2x} dx = \frac{3}{8}e$$

(4)

- (b) Use the substitution $u = \tan x$ to find the exact value of

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x} dx$$

(8)

(Total 12 marks)

Q13.

- (a) Find $\int x\sqrt{x^2 + 3} dx$.

(2)

- (b) Solve the differential equation

$$\frac{dy}{dx} = \frac{x\sqrt{x^2 + 3}}{e^{2y}}$$

given that $y = 0$ when $x = 1$. Give your answer in the form $y = f(x)$.

(7)

(Total 9 marks)

Q14.

A curve has equation $y = x^5 - 2x^2 + 9$. The point P with coordinates $(-1, 6)$ lies on the curve.

- (a) Find the equation of the tangent to the curve at the point P , giving your answer in

the form $y = mx + c$.

(5)

(b) The point Q with coordinates $(2, k)$ lies on the curve.

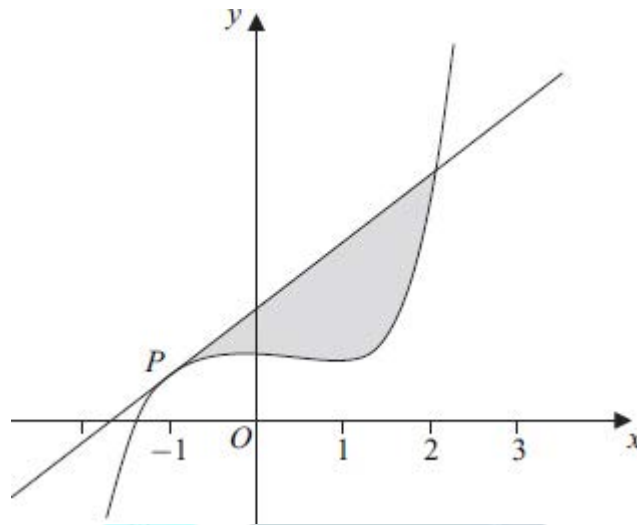
(i) Find the value of k .

(1)

(ii) Verify that Q also lies on the tangent to the curve at the point P .

(1)

(c) The curve and the tangent to the curve at P are sketched below.



(i) Find $\int_{-1}^2 (x^5 - 2x^2 + 9) dx$.

(5)

(ii) Hence find the area of the shaded region bounded by the curve and the tangent to the curve at P .

(3)

(Total 15 marks)

Q15.

(a) (i) Using the binomial expansion, or otherwise, express $(2 + y)^3$ in the form $a + by + cy^2 + y^3$, where a , b and c are integers.

(2)

(ii) Hence show that $(2 + x^{-2})^3 + (2 - x^{-2})^3$ can be expressed in the form $p + qx^{-4}$, where p and q are integers.

(3)

(b) (i) Hence find $\int [(2 + x^{-2})^3 + (2 - x^{-2})^3] dx$.

(2)

(ii) Hence find the value of $\int_1^2 \left[(2+x^{-2})^3 + (2-x^{-2})^3 \right] dx$.

(2)

(Total 9 marks)

Q16.

(a) (i) Express $\frac{5-8x}{(2+x)(1-3x)}$ in the form $\frac{A}{2+x} + \frac{B}{1-3x}$, where A and B are integers.

(3)

(ii) Hence show that $\int_{-1}^0 \frac{5-8x}{(2+x)(1-3x)} dx = p \ln 2$, where p is rational.

(4)

(b) (i) Given that $\frac{9-18x-6x^2}{2-5x-3x^2}$ can be written as $C + \frac{5-8x}{2-5x-3x^2}$, find the value of C .

(1)

(ii) Hence find the exact value of the area of the region bounded by the curve $y = \frac{9-18x-6x^2}{2-5x-3x^2}$, the x -axis and the lines $x = -1$ and $x = 0$.

You may assume that $y > 0$ when $-1 \leq x \leq 0$.

(2)

(Total 10 marks)

Q17.

(a) Find $\int t \cos\left(\frac{\pi}{4}t\right) dt$.

(4)

(b) The platform of a theme park ride oscillates vertically. For the first 75 seconds of the ride,

$$\frac{dx}{dt} = \frac{t \cos\left(\frac{\pi}{4}t\right)}{32x}$$

where x metres is the height of the platform above the ground after time t seconds.

At $t = 0$, the height of the platform above the ground is 4 metres.

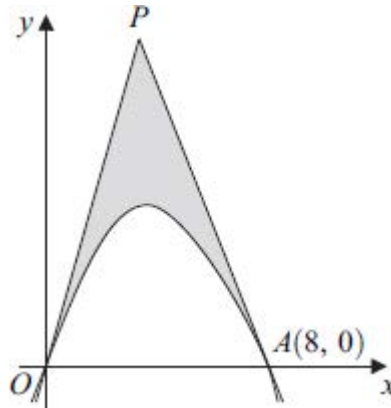
Find the height of the platform after 45 seconds, giving your answer to the nearest

centimetre.

(6)
(Total 10 marks)

Q18.

The diagram shows part of a curve crossing the x -axis at the origin O and at the point $A(8, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve has equation

$$y = 12x - 3x^{\frac{5}{3}}$$

- (a) Find $\frac{dy}{dx}$. (2)

- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)

- (ii) Show that the equation of the tangent at $A(8, 0)$ is $y + 8x = 64$. (3)

- (c) Find $\int \left(12x - 3x^{\frac{5}{3}} \right) dx$. (3)

- (d) Calculate the area of the shaded region bounded by the curve from O to A and the tangents OP and AP . (7)
- (Total 17 marks)

Q19.

- (a) Express $\frac{2x+3}{4x^2-1}$ in the form $\frac{A}{2x-1} + \frac{B}{2x+1}$, where A and B are integers. (3)
- (b) Express $\frac{12x^3-7x-6}{4x^2-1}$ in the form $Cx + \frac{D(2x+3)}{4x^2-1}$, where C and D are integers. (3)
- (c) Evaluate $\int_1^2 \frac{12x^3-7x-6}{4x^2-1} dx$, giving your answer in the form $p + \ln q$, where p and q are rational numbers. (5)
- (Total 11 marks)

Q20.

Solve the differential equation

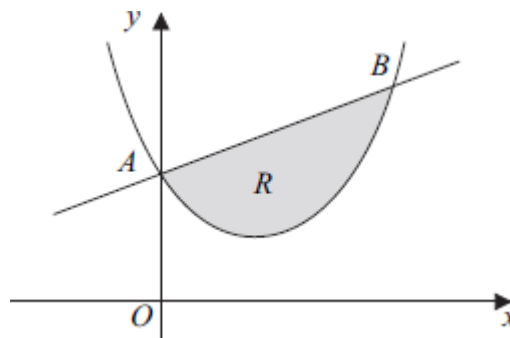
$$\frac{dy}{dx} = y^2 x \sin 3x$$

given that $y = 1$ when $x = \frac{\pi}{6}$. Give your answer in the form $y = \frac{9}{f(x)}$.

(Total 9 marks)

Q21.

- (a) (i) Express $x^2 - 3x + 5$ in the form $(x-p)^2 + q$. (2)
- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = x^2 - 3x + 5$. (1)
- (b) The curve C with equation $y = x^2 - 3x + 5$ and the straight line $y = x + 5$ intersect at the point $A(0, 5)$ and at the point B , as shown in the diagram below.



- (i) Find the coordinates of the point B .

(3)

- (ii) Find $\int (x^2 - 3x + 5) dx$.

(3)

- (iii) Find the area of the shaded region R bounded by the curve C and the line segment AB .

(4)

(Total 13 marks)

Q22.

- (a) Expand $\left(x^{\frac{3}{2}} - 1\right)^2$.

(2)

- (b) Hence find $\int \left(x^{\frac{3}{2}} - 1\right)^2 dx$.

(3)

- (c) Hence find the value of $\int_1^4 \left(x^{\frac{3}{2}} - 1\right)^2 dx$.

(2)

(Total 7 marks)

Q23.

At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^2 - \frac{4}{x^2} - 11$$

The point $P(2, 1)$ lies on the curve.

- (a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 2$.

(1)

- (ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 2$.

(4)

- (iii) Hence state whether P is a maximum point or a minimum point, giving a reason for your answer.

(1)

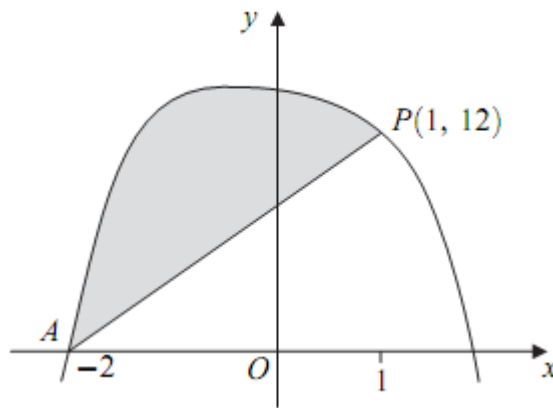
- (b) Find the equation of the curve.

(4)

(Total 10 marks)

Q24.

The curve sketched below passes through the point $A(-2, 0)$.



The curve has equation $y = 14 - x - x^4$ and the point $P(1, 12)$ lies on the curve.

- (a) (i) Find the gradient of the curve at the point P .

(3)

- (ii) Hence find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$.

(2)

- (b) (i) Find $\int_{-2}^1 (14 - x - x^4) dx$.

(5)

- (ii) Hence find the area of the shaded region bounded by the curve $y = 14 - x - x^4$ and the line AP .

(2)

(Total 12 marks)

Q25.

- (a) Using the binomial expansion, or otherwise, express $(1 - x)^3$ in ascending powers of x .

(2)

- (b) Show that the expansion of

$$(1 + y)^4 - (1 - y)^3$$

is

$$7y + py^2 + qy^3 + y^4$$

where p and q are constants to be found.

(4)

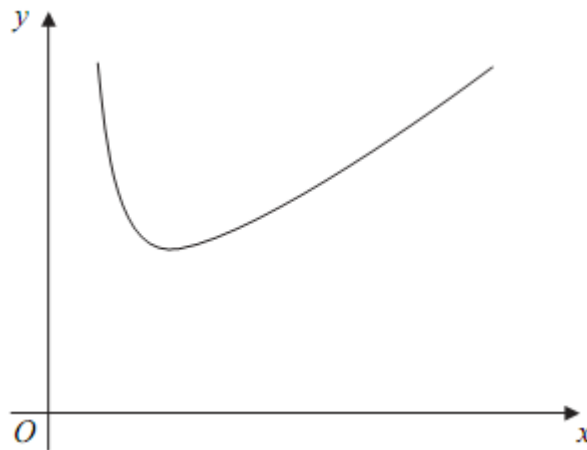
- (c) Hence find $\int \left[(1 + \sqrt{x})^4 - (1 - \sqrt{x})^3 \right] dx$, expressing each coefficient in its simplest form.

(4)

(Total 10 marks)

Q26.

A curve C is defined for $x > 0$ by the equation $y = x + 3 + \frac{8}{x^4}$ and is sketched below.



- (a) Given that $y = x + 3 + \frac{8}{x^4}$, find $\frac{dy}{dx}$.

(3)

- (b) Find an equation of the tangent at the point on the curve C where $x = 1$.

(3)

- (c) The curve C has a minimum point M . Find the coordinates of M .

(4)

- (d) (i) Find $\int \left(x + 3 + \frac{8}{x^4} \right) dx$.

(3)

- (ii) Hence find the area of the region bounded by the curve C , the x -axis and the lines $x = 1$ and $x = 2$.

(2)

- (e) The curve C is translated by $\begin{bmatrix} 0 \\ k \end{bmatrix}$ to give the curve $y = f(x)$. Given that the x -axis is a tangent to the curve $y = f(x)$, state the value of the constant k .

(1)

(Total 16 marks)

Q27.

- (a) Find $\int \frac{1}{3+2x} dx$.

(2)

- (b) By using integration by parts, find $\int x \sin \frac{x}{2} dx$.

(4)

(Total 6 marks)

Q28.

- (a) (i) Solve the differential equation $\frac{dx}{dt} = \sqrt{x} \sin\left(\frac{t}{2}\right)$ to find x in terms of t .

(3)

- (ii) Given that $x = 1$ when $t = 0$, show that the solution can be written as

$$x = (a - \cos bt)^2$$

where a and b are constants to be found.

(3)

- (b) The height, x metres, above the ground of a car in a fairground ride at time t seconds is

modelled by the differential equation $\frac{dx}{dt} = \sqrt{x} \sin\left(\frac{t}{2}\right)$.

The car is 1 metre above the ground when $t = 0$.

- (i) Find the greatest height above the ground reached by the car during the ride.

(2)

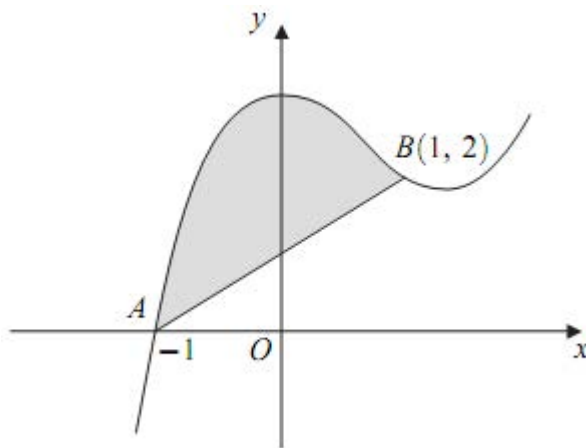
- (ii) Find the value of t when the car is first 5 metres above the ground, giving your answer to one decimal place.

(2)

(Total 10 marks)

Q29.

The curve with equation $y = x^3 - 2x^2 + 3$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and passes through the point $B(1, 2)$.

- (a) Find $\int_{-1}^1 (x^3 - 2x^2 + 3) dx$. (5)
- (b) Hence find the area of the shaded region bounded by the curve $y = x^3 - 2x^2 + 3$ and the line AB . (3)
- (Total 8 marks)

Q30.

- (a) The expression $(2 + x^2)^3$ can be written in the form

$$8 + px^2 + qx^4 + x^6$$

Show that $p = 12$ and find the value of the integer q .

(3)

- (b) (i) Hence find $\int \frac{(2 + x^2)^3}{x^4} dx$. (5)

- (ii) Hence find the exact value of $\int_1^2 \frac{(2 + x^2)^3}{x^4} dx$.

(2)

(Total 10 marks)

Q31.

- (a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

- (i) Show that $\frac{dy}{dx} = \frac{1}{k \cos \theta}$, where k is an integer. (4)

- (ii) Find an equation of the normal to the curve at the point where $\theta = \frac{\pi}{3}$. (4)

- (b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$. (5)

(Total 13 marks)

Q32.

At the point (x, y) on a curve, where $x > 0$, the gradient is given by

$$\frac{dy}{dx} = 7\sqrt{x^5} - 4$$

- (a) Write $\sqrt{x^5}$ in the form x^k , where k is a fraction. (1)

- (b) Find $\int (7\sqrt{x^5} - 4) dx$. (3)

- (c) Hence find the equation of the curve, given that the curve passes through the point $(1, 3)$. (3)

(Total 7 marks)

Q33.

The expression $\frac{10x^2 + 8}{(x+1)(5x-1)}$ can be written in the form $2 + \frac{A}{x+1} + \frac{B}{5x-1}$, where A and B are constants.

- (a) Find the values of A and B . (4)

- (b) Hence find $\int \frac{10x^2 + 8}{(x+1)(5x-1)} dx$. (4)

(Total 8 marks)

Q34.

Solve the differential equation $\frac{dy}{dx} = \frac{1}{y} \cos\left(\frac{x}{3}\right)$, given that $y = 1$ when $x = \frac{\pi}{2}$.

Write your answer in the form $y^2 = f(x)$.

(Total 6 marks)

Q35.

(a) The expression $\left(1 - \frac{1}{x^2}\right)^3$ can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} - \frac{1}{x^6}$$

Find the values of the integers p and q .

(2)

(b) (i) Hence find $\int \left(1 - \frac{1}{x^2}\right)^3 dx$.

(4)

(ii) Hence find the value of $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx$.

(2)

(Total 8 marks)

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